

Skills Summary

Turnaround and Break-Apart Fact Strategies

Use this reference while completing your worksheet or when studying.

Skill 1 — Turnaround Facts

Rule — the order of the factors does not change the product. If you know one fact, you know its turnaround for free. Picture an array of counters: turning the array on its side swaps the rows and columns but does not add or remove a single counter.

Example: You know $6 \times 3 = 18$. What is 3×6 ?

Step 1. The same two factors are being multiplied, just written in the other order, so no new counting is needed.

Step 2. Turn the fact around: 3×6 has the same product as 6×3 .

$\Rightarrow 18$

Try it: You know $5 \times 4 = 20$. What is 4×5 ?

check: 20

Skill 2 — Breaking a Factor Apart

Rule — split one factor, multiply each piece, then put the pieces together. Add the two part-products when you split by adding, and take one away when you split by subtracting:

- Split by adding: $6 \times 8 = (6 \times 5) + (6 \times 3)$
- Split by subtracting: $9 \times 7 = (10 \times 7) - 7$

Split off a five or a ten — those are the pieces you can already count.

Example: Multiply 6×8 by breaking the 8 into $5 + 3$.

Step 1. Eights are hard to count by, but fives and threes are easy, so split the harder factor into two friendly pieces.

Step 2. $6 \times 5 = 30$ and $6 \times 3 = 18$, and the two part-products go back together by adding.

$\Rightarrow 48$

Try it: Use $9 \times 7 = (10 \times 7) - 7$ to find 9×7 .

check: 63

(!) Watch out: when you go down from a ten, take away a whole *group*, not one. For 9×7 you remove one group of seven, so the step is $70 - 7$.

Skill 3 — Dividing with a Missing-Factor Fact

Rule — every division fact is a multiplication fact with a hole in it. To work out a division, ask what fills the hole in the matching multiplication fact, then use the times table you already know.

Example: What is $18 \div 3$?

Step 1. Dividing by three asks how many threes make eighteen, which is the same as filling the hole in a multiplication fact.

Step 2. Count the threes: 3, 6, 9, 12, 15, 18 — that is six of them, so the missing factor is six.
 $\Rightarrow \quad \mathbf{n = 6}$

Try it: What is $28 \div 4$? Think: $4 \times \underline{\quad} = 28$.

check: $4 \times 7 = 28$